

Shaik Roshan
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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2,BL3)

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10

Total Marks: 50

Annexure A

Case Study

Duration: 2hrs

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. **(5 Marks)**
2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. **(5 Marks)**

3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. **(5 Marks)**
4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. **(5 Marks)**
5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. **(5 Marks)**
6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. **(5 Marks)**
7. A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance. **(5 Marks)**
8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. **(5 Marks)**
9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. **(5 Marks)**

10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. **(5 Marks)**

Code of Conduct:

I S. Roshan (19242558) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CASE STUDY

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured , and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

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Regno: 192425258

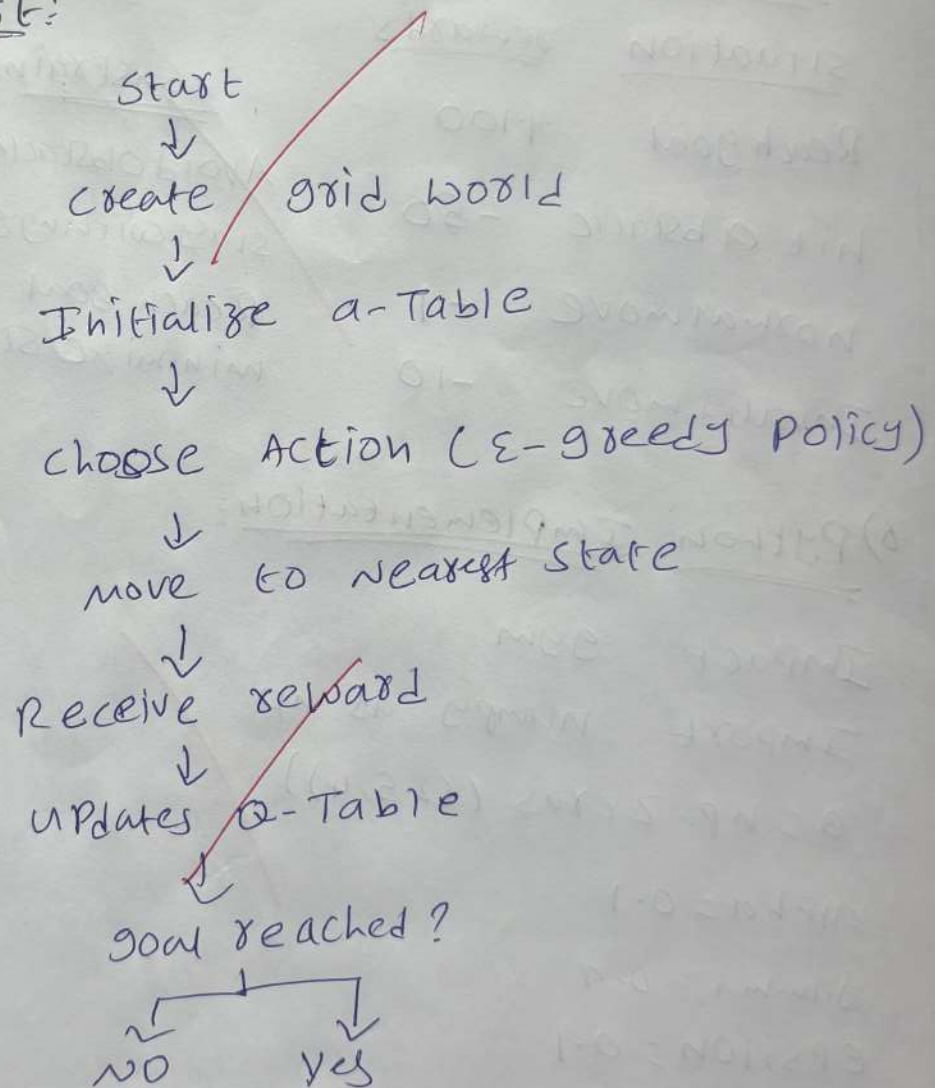
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Develop a Python based Reinforcement learning model using open AI gym to gold world navigation

1) Problem understanding:

The objective is to develop a (RL) agent that navigates through a grid world environment.

2) Flow Chart:



3) Reinforcement learning components

components

environment
Agent
state space
action space
goal state
obstacle state

description

5x5 grid world
RL Agent
each grid cell represents one state
up, down, left, right
Reach destination safely

4) Reward function

<u>situation</u>	<u>rewards</u>
Reach goal	+100
hit obstacle	-50
normal move	-1
Invalid move	-10

5) constraints

<u>constraint</u>	<u>purpose</u>
Avoid obstacles	collision
stay within goals	valid
Reach goal	navigation
minimize steps	efficiency

6) Python Implementation:

```
Impact gym
Import numpy as np
a = np.zeros (25, 4)
alpha = 0.1
gamma = 0.9
epsilon = 0.1
```

7) Working Process:

- 1) Create the grid world environment
- 2) Initialize the a-table with zero values.

- 5) Receive reward from agent reaches the goal
- 6) store the learned policy for navigation

7) Belman equation:

$$Q(s,a) = Q(s,a) + \alpha (R + \gamma \max_{a'} Q(s',a') - Q(s,a))$$

8) Conclusion:

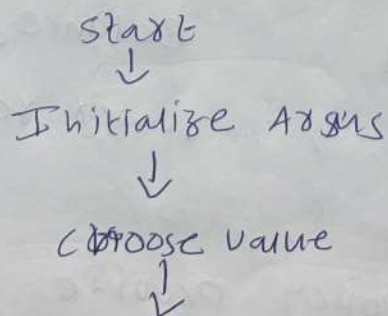
The Q-learning based (RL) model successfully learns an optimal navigation policy.

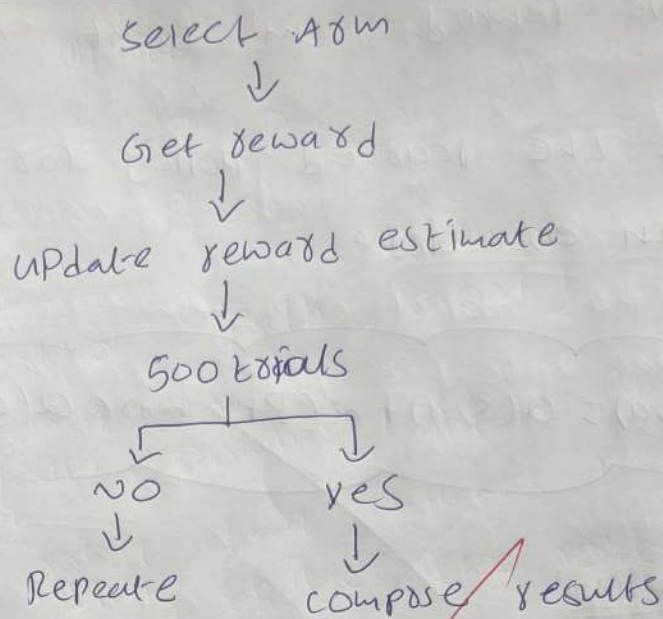
9) Greedy multi-Armed Bandit Algorithm:

Problem statement:

The ϵ -greedy algorithm helps an agent maximize rewards by balancing exploration within 500 trials

Flow chart:





Parameters:

<u>Parameters</u>	<u>value</u>
Arms	5
Trials	500
epsilon	0.1, 0.3, 0.5

Comparison:

<u>ϵ</u>	<u>exploration</u>	<u>Performance</u>
0.1	low	Best reward
0.3	reward	Balance
0.5	high	lower reward

Conclusion:

The $\epsilon=0.1$ policy provide the highest cumulative rewards by balancing omitted exploration

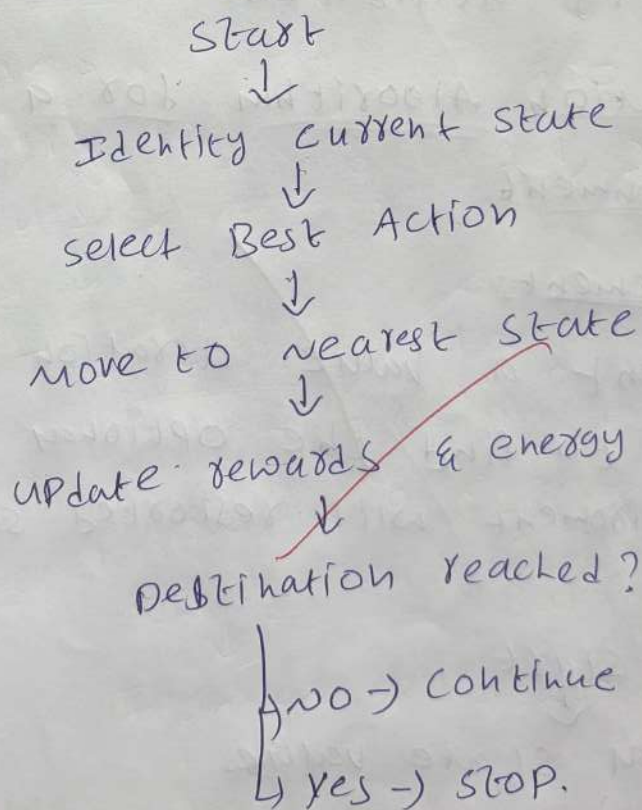
With maximize exploration under the 500 trial constraint.

3) Design a Markov Decision Process (MDP) for an Autonomous Robot:

Problem statement:

Design an MDP model for an autonomous robot that reaches its destination while reducing energy usage and avoiding unnecessary movement.

Flow chart:



Python Syntax:

state = ["start", "path", "goal"]

actions = ["up", "down", "left", "right"]

rewards = 100

battery-limit = 30.

Conclusion:

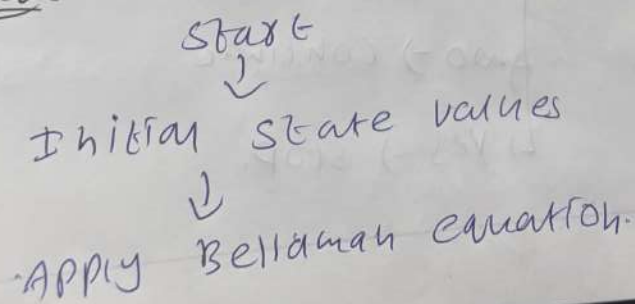
The MDP enables the robot to choose the optimal path while making energy consumption improving navigation efficiency and battery utilization.

4) Value Iteration Algorithm for a constrained grid environment

Problem Statement:

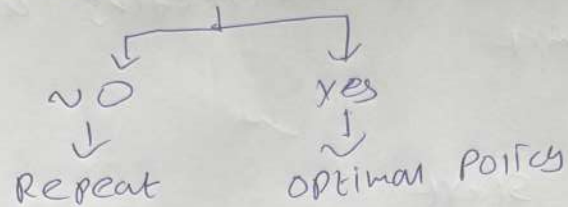
Implement a value iteration algorithm using python to find the optimal policy in a grid environment with restricted states.

Flowchart:



Update state values

↓
check convergence



Bellman Equation

$$V(s) = \max_a [R + \gamma V(s')]]$$

Python Syntax:

import tensorflow as tf

import numpy as np

gamma = 0.9

values = np.zeros((5, 5))

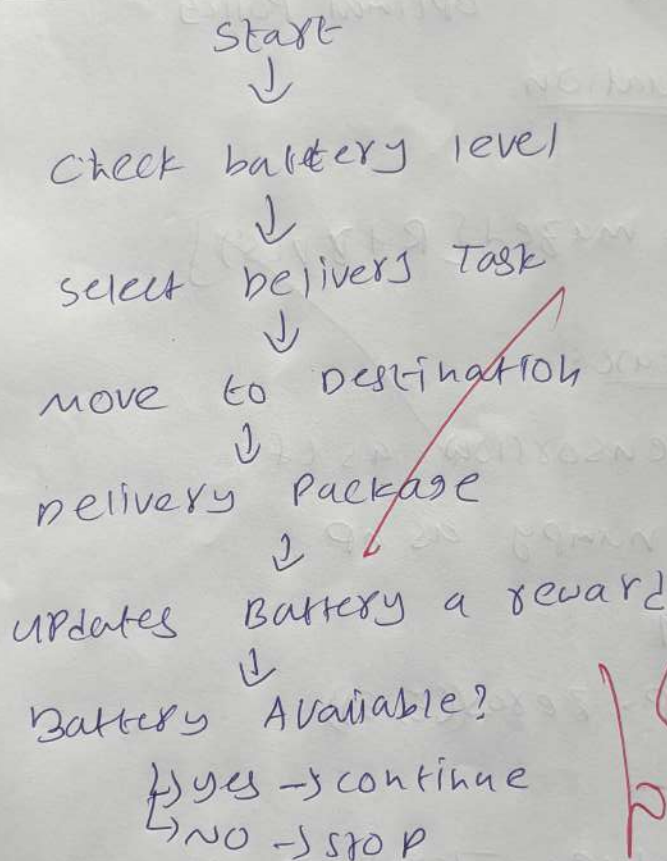
Conclusion:

The value Iteration algorithm applies the Bellman equation to find the optimal policy while avoiding restricted states and maximizing long term rewards.

5) RL Framework for a warehouse robot with
Battery constraint
Problem statement:

Design a (RL) frame work for a warehouse robot that completes the maximum no of deliveries without exceeding a 30-minute Battery Limit

Flowchart:



Python syntax:

```
battery = 30  
reward = 50  
if battery > 0:  
    action = 'delivers'
```

Conclusion:

The RL frame work helps the warehouse robot maximize deliveries while staying within the battery.